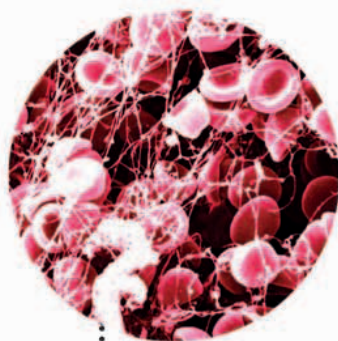




▼ **1663** Italian doctor *Marcello Malpighi* discovers blood capillaries, helping to confirm that blood circulates around the body.



▲ **1674–77** Dutchman *Antonie van Leeuwenhoek* discovers microscopic red blood cells, sperm, and bacteria.



1691 English doctor *Clopton Havers* describes the structure of bones.

▼ **1775** French chemist *Antoine Lavoisier* discovers oxygen, and later shows that cell respiration is, like burning, a chemical process that consumes oxygen.



▲ **1800** French doctor *Marie François Bichat* publishes a book in which he shows that organs are made of different groups of cells called tissues.

► **1816** The stethoscope, a device used to listen to a patient's heart and breathing sounds, is invented by French doctor *René Laënnec*.



1650

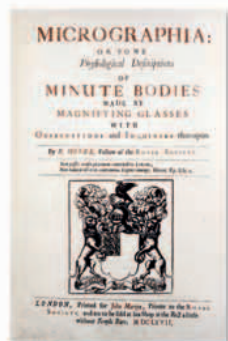
1700

1800

1835

1667 English doctor *Richard Lower* carries out a blood transfusion from a sheep to the student *Arthur Coga*. Amazingly, Coga survives.

▼ **1665** English scientist *Robert Hooke* publishes *Micrographia*, in which he uses the new term "cell."



1672 *Regnier de Graaf*, a Dutch doctor, describes the structure and workings of the female reproductive system.



▲ **1780** Italian doctor *Luigi Galvani* experiments with nerves, muscles, and electricity.

▲ **1747** British naval doctor *James Lind* discovers that citrus fruits prevent the vitamin C deficiency disease scurvy during long sea voyages.

1833 American surgeon *William Beaumont* records the results of his researches into the mechanism of digestion.

▼ **1796** The first vaccination against smallpox is carried out on an eight-year-old boy by English doctor *Edward Jenner*.

